

WASP-1b

R-1536

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-1_130910 · created 2026-08-01T18:44:16+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456545.8677 +/- 0.0053 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.139 +/- 0.039
Transit depth (Rp/Rs)^2	1.93 +/- 1.08 [%]
Orbital Inclination (inc)	89.6 +/- 1.43
Airmass coefficient 1 (a1)	3.23 +/- 1.81
Airmass coefficient 2 (a2)	-1.16 +/- 1.18
Scatter in the residuals of the lightcurve fit is	51.2 %
Best Comparison Star	None
Optimal Aperture	2.77
Optimal Annulus	5.77
Transit Duration (day)	0.157 +/- 0.0055

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.139 ± 0.039 vs 0.1036 (deviation 0.91σ)
Duration fit vs published	0.157 d vs 0.1567 d (deviation 0.06σ)
Mid-time O–C	12.6 min
Depth SNR	3.6
Residual scatter	51.2 %

Light curve

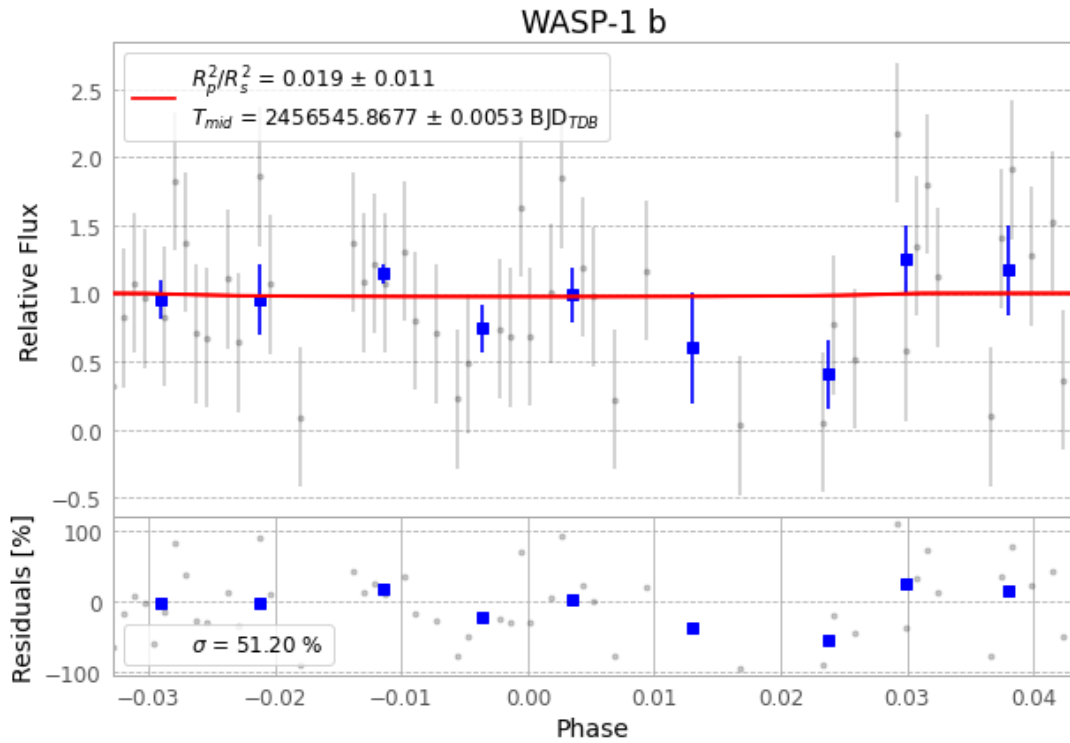


Figure 1 — FinalLightCurve_WASP-1 b_2013-09-09.png (SHA-256 7350165af6f341ba...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [275, 238], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3474472981e27918...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

94 FITS frames (62 MB), WASP-1130910064212.FITS → WASP-1130910112712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-1-130910.log (8c4007d530ef...), FinalLightCurve_WASP-1 b_2013-09-09.png (7350165af6f3...), FinalLightCurve_WASP-1 b_2013-09-09.csv (5b50b0761a9c...)

Compute resources

Wall time 200.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 18:44Z.